

Position-wise
FFN



$$w_2^T (\text{Relu}(w_1^T a_2 + b_1)) + b_2$$

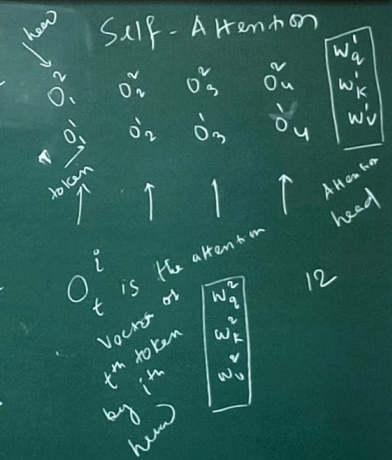
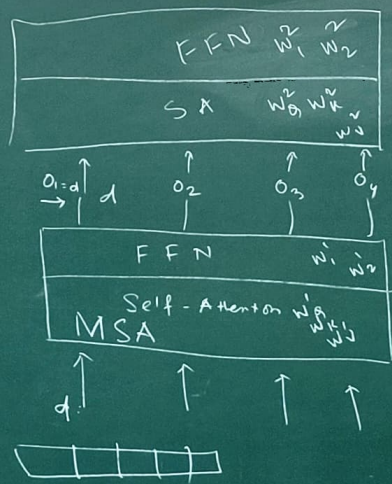
$$w_2^T (\text{Relu}(w_1^T a_1 + b_1)) + b_2$$

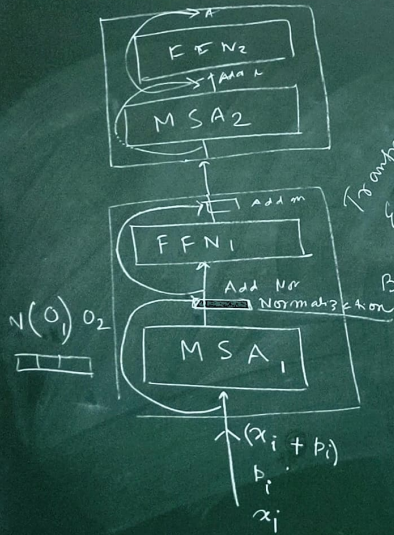
Activation
Attention o/p

Self-Attention block

$$d \rightarrow d/2 \rightarrow d$$

$$w_1 \mid d \times d/2 \quad w_2 \mid d/2 \times 1$$

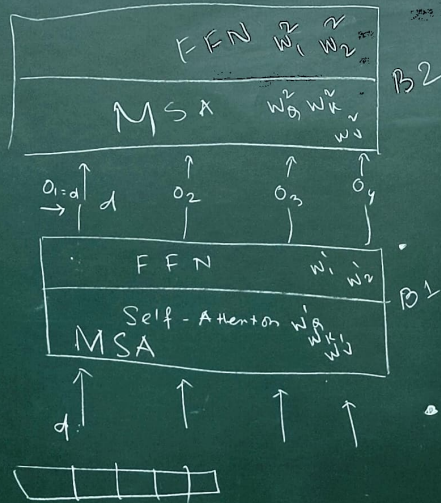
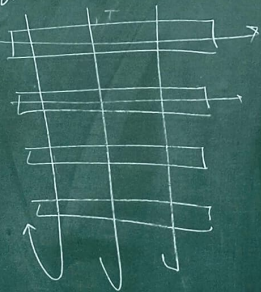




Transformer Encoder

Min-Max

$\frac{x_i - \bar{x}}{\sigma}$



Self-Attention

tokens

o_1, o_2, o_3, o_4

o_i is the attention

Vector of tokens by in head

Attention head

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